

Logarithmic Functions & Models

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **convert** between logarithmic and exponential forms.

RELATED SKILLS

- I Convert between logarithmic and exponential forms.
- I Interpret a logarithm as an exponent.

☐ 2. I can **evaluate** a logarithm using its exponential meaning.

RELATED SKILLS

- I Rewrite exponential expressions using a common base when possible.
- D Interpret a logarithm as an exponent.

☐ 3. I can **expand or condense** an expression using product, quotient, and power properties.

RELATED SKILLS

- I Apply the logarithm power property in either direction.
- I Apply the logarithm product property in either direction.
- I Apply the logarithm quotient property in either direction.
- A Simplify an algebraic expression using operation properties.

☐ 4. I can **solve** an exponential equation using logarithms.

RELATED SKILLS

- I Isolate an exponential expression and apply a logarithm.
- A Apply the logarithm power property in either direction.
- A Isolate a specified variable in an equation or formula.

☐ 5. I can **solve** a logarithmic equation using exponential form and check domain restrictions.

RELATED SKILLS

- I Combine or isolate logarithms before converting to exponential form.
- I Reject solutions that make a logarithm argument nonpositive.
- A Convert between logarithmic and exponential forms.

☐ 6. I can **calculate and interpret** doubling time or half-life.

RELATED SKILLS

- I Translate a doubling-time or half-life question into an exponential equation.
- D Explain a model parameter in the language of its context.
- A Attach and interpret units for rates, inputs, and outputs.
- A Isolate an exponential expression and apply a logarithm.

☐ 7. I can **determine** domain, range, intercepts, and asymptotes of a logarithmic function.

RELATED SKILLS

- I Identify vertical asymptotic behavior of a logarithmic graph.
- D Connect parameters in a formula to features of its graph.
- D Determine domain restrictions from denominators, even roots, and logarithms.
- A Reject solutions that make a logarithm argument nonpositive.

- ☐ **8. I can **explain** how exponential and logarithmic functions undo one another.**

RELATED SKILLS

- D** Interpret a logarithm as an exponent.
- A** Exchange input and output values when interpreting an inverse.
- A** Verify inverse functions using composition.

- ☐ **9. I can **solve** a multi-step exponential or logarithmic equation.**

RELATED SKILLS

- D** Combine or isolate logarithms before converting to exponential form.
- D** Isolate an exponential expression and apply a logarithm.
- A** Apply the logarithm power property in either direction.
- A** Apply the logarithm product property in either direction.
- A** Apply the logarithm quotient property in either direction.
- A** Reject solutions that make a logarithm argument nonpositive.

- ☐ **10. I can **determine** an intersection of exponential functions graphically or algebraically.**

RELATED SKILLS

- D** Isolate an exponential expression and apply a logarithm.
- A** Check an ordered pair in both equations of a system.
- A** Compare exponential growth rates expressed in different forms.
- A** Locate and interpret the intersection of two lines.

- ☐ **11. I can **rewrite** a discrete exponential model as an equivalent continuous exponential model.**

RELATED SKILLS

- D** Convert between equivalent discrete- and continuous-growth forms.
- A** Convert a percent increase or decrease to a growth or decay factor.
- A** Isolate an exponential expression and apply a logarithm.

- ☐ **12. I can **solve and interpret** an exponential or logarithmic modeling problem.**

RELATED SKILLS

- D** Explain a model parameter in the language of its context.
- D** Identify initial amount, rate, and time interval in context.
- A** Attach and interpret units for rates, inputs, and outputs.
- A** Combine or isolate logarithms before converting to exponential form.
- A** Isolate an exponential expression and apply a logarithm.
- A** Restrict a model to meaningful contextual inputs.